Dinusha Madushanka • Final Year Project • BSc (Hons) Computing

AI-Powered Restaurant Management System

A Real-Time, Multi-Branch Cloud Platform Replacing Fragmented POS, Kitchen & Order Workflows

🎯 Aims & Objectives

- **Goal:** Unify POS, kitchen, waiter, inventory, finance & AI analytics in one real-time platform
- Kitchen tickets delivered to the correct station in under 1 second
- Branch- and role-level access enforced at the database layer
- Usable through short connectivity outages; sales linked directly to stock via BOM deduction

🔍 Rationale & Context

Problem: Multi-branch restaurants on paper tickets and standalone POS terminals suffer fragmented order visibility, no cross-branch oversight, and stock that silently drifts from actual sales.

Key Pain Points: No shared realtime order record · 10 staff roles need scoped access · sales rarely trigger stock deduction

📚 Literature Review

- Commercial platforms rarely combine branch-scoped RBAC, offline-first PWAs and finance in one architecture
- Trigger-driven realtime beats polling for sub-second updates
- DB-enforced Row Level Security is safer than app-only checks
- Offline queue-and-replay is standard for unreliable connectivity

🗺️ Scope & Data Model

- 7 client layouts (ERP, waiter/manager/delivery PWAs, kiosk, customer menu, auth) share one schema
- Core entities: branches, staff/shifts, orders, order items, kitchen tickets, inventory, ledgers
- Every table carries a `branch_id` for tenant isolation; BOM links items to stock

🚧 Challenges & Solutions

Trigger Ordering
Kitchen assignment must precede status change or KOTs lose routing

Cross-Branch RLS Leak
Unscoped policies fixed & re-tested

Non-Atomic Billing
3-step billing merged into one atomic RPC

Dual Realtime Systems
Two layers flagged as debt to consolidate

⚙️ Technical Implementation

```
Vue 3 + Quasar 2 (SPA / PWA / Electron / Tauri)
                    ⇅
Supabase (PostgREST · Auth · Realtime)
                    ⇅
PostgreSQL 17 — RLS · Triggers · RPCs
```

Vue 3 · Quasar 2 · Supabase · PostgreSQL 17 · Pinia · Electron/Tauri · Gemini

- Trigger-driven Kitchen Order Ticket (KOT) generation, routed per station
- Realtime pipeline via PostgreSQL logical replication
- Offline-first sync (IndexedDB queue & replay)
- 10-role RBAC enforced by database Row Level Security
- QR-code table ordering & touch-menu kiosk flow

🧪 Methodology

Iterative, feature-driven build against a live backend — module by module: auth/RBAC → POS & orders → KOT/kitchen display → waiter PWA → QR ordering → inventory → finance → HR/payroll → AI dashboard.

- Automated unit/integration tests (Vitest) for RBAC & store logic
- Manual end-to-end walkthroughs of the order lifecycle
- Dedicated security review of RLS policies

📊 Key Results

<1s	10	13
Order→Kitchen Latency	Roles Under Branch RBAC	Defects Found & Fixed

Reliability: Offline sync preserves order data through short outages, replaying queued writes in order on reconnect.

⚡ Evaluation Highlights

- Cross-branch data isolation confirmed via targeted RLS penetration checks
- Client-side route guards verified as UX-only — DB is the real security boundary
- BOM stock deduction functionally complete; recipe coverage remains partial

🔮 Future Enhancements

- **Migration Tooling:** Formal, versioned schema migrations
- **Realtime Consolidation:** Merge dual subscription layers
- **BOM Coverage:** Complete recipe data across the menu
- **AI Forecasting:** Benchmark accuracy against real sales data

🎓 Academic Contributions

- Worked example combining trigger-driven realtime, DB-enforced multi-tenancy and offline-first PWA design
- Critical evaluation of theory-to-practice gaps in RLS composition and realtime architecture

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